347631

Course Name: **Digital Image Processing**

Course Outcome

CO1- Understand mathematical formulation of an image, its processing steps and relationship between image pixels.

CO2- Apply Image enhancement using intensity transformations and spatial filtering.

CO3- Analyze image enhancement for frequency domain using Fourier transform.

CO4- Formulate region of interest through morphological operations.

CO5- Evaluate strongly co-related regions obtained through Segmentation using discontinuity and homogeneity-based segmentation techniques

CO6: Describe an object of an image using Shape Number and Boundary descriptors.

University Roll No. ....

Printed Pages :

Mid Term Examination, Odd Semester 2024-25

B.Tech. (CSE/IIOT), III Year, V Sem

Subject Code & Subject Name- BCSE 0101 & Digital Image Processing

Maximum Marks: 30

Time: 2 Hours

Instruction for students:

1. All parts of a question should be answer at one place.
2. First complete the one set of section then start a new section.

Section – A

3 X 5 = 15 Marks

Attempt All Questions

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the image segment as shown. Let V= {0,1}, compute the length of the shortest 4 path, 8 path and m path between the pixels p and q. If a particular path does not exist between these two points, explain why? <table><tr><td>3</td><td>1</td><td>2</td><td>1:q</td></tr><tr><td>2</td><td>2</td><td>0</td><td>2</td></tr><tr><td>1</td><td>2</td><td>1</td><td>1</td></tr><tr><td>p:1</td><td>0</td><td>1</td><td>2</td></tr></table>	3	1	E	C
2	A gray level image in the range from 10-60. If we generate a linear contrast stretched image with minimum and maximum gray level are 120 and 180, then how the new intensity gets mapped. Write formula and compute the gray level transformation. <table><tr><td>10</td><td>15</td></tr><tr><td>20</td><td>50</td></tr></table>	3	2	An	C
3	A student has a grayscale image of a landscape where the darker areas represent shadows and the lighter areas represent bright regions. The student wants to apply an intensity transformation on the image effectively to	3	2	An	M

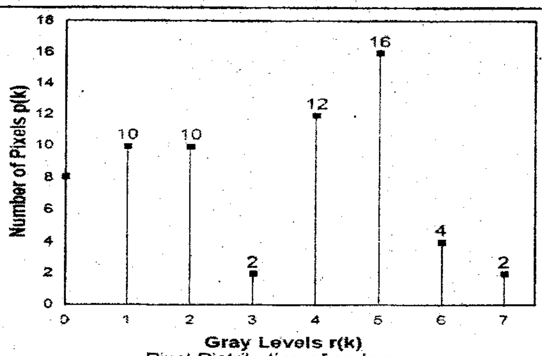
<table>
<tr><td></td><td>reverse the intensity levels so that the shadows become bright and the bright regions become dark. Explain which transform should he use. Apply the same transformation on the given image.<table><tr><td>122</td><td>150</td><td>200</td></tr><tr><td>225</td><td>225</td><td>225</td></tr><tr><td>250</td><td>250</td><td>240</td></tr></table></td><td></td><td></td><td></td><td></td></tr>
<tr><td>4</td><td>A common measure of transmission for digital data is the baud rate, defined as symbols (bits in our case) per second. As a minimum, transmission is accomplished in packets consisting of 14 start bits, a byte (8 bit) of information and 14 stop bits. Using these facts, answer the following:
(i) How many seconds would it take to transmit a sequence of 200 images of size 1280 X 960 pixels with 256 intensity levels using a 3 M-baud (10^6 bits/sec) modem? (This is a representative medium speed for digital subscriber line residential medium.)
(ii) What would be the time using a 30 G-baud (10^6 bits/sec) modem?</td><td>3</td><td>1</td><td>U</td><td>C</td></tr>
<tr><td>5</td><td>What are sharpening spatial filters? Discuss the response of first and second order derivatives. What is the Laplacian mask for image sharpening?</td><td>3</td><td>2</td><td>An</td><td>C</td></tr>
</table>

Section – B

Attempt All Questions 5 X 3 = 15

Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Explain the steps in image processing with the help of block diagram. OR Discuss the different types of basic intensity transformation functions.	5	1	U	P
7	Perform histogram equalization for the 8 X 8, eight level image described in Figure. Also, construct the histogram of the equalized image.	5	2	C	P

<table>
<tr><td></td><td>

Pixel Distribution of an Image</td><td></td><td></td><td></td><td></td></tr>
<tr><td>8</td><td>Consider the following 5-bit, 4 X 4 image
(i) Extract the 2nd Bit Plane.
(ii) How will image look in binary, if thresholding is set at 27?
<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td></tr>
<tr><td>16</td><td>26</td><td>25</td><td>21</td></tr>
<tr><td>24</td><td>25</td><td>26</td><td>16</td></tr>
<tr><td>28</td><td>29</td><td>30</td><td>31</td></tr>
</table></td><td>5</td><td>2</td><td>U</td><td>C</td></tr>
</table>

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities